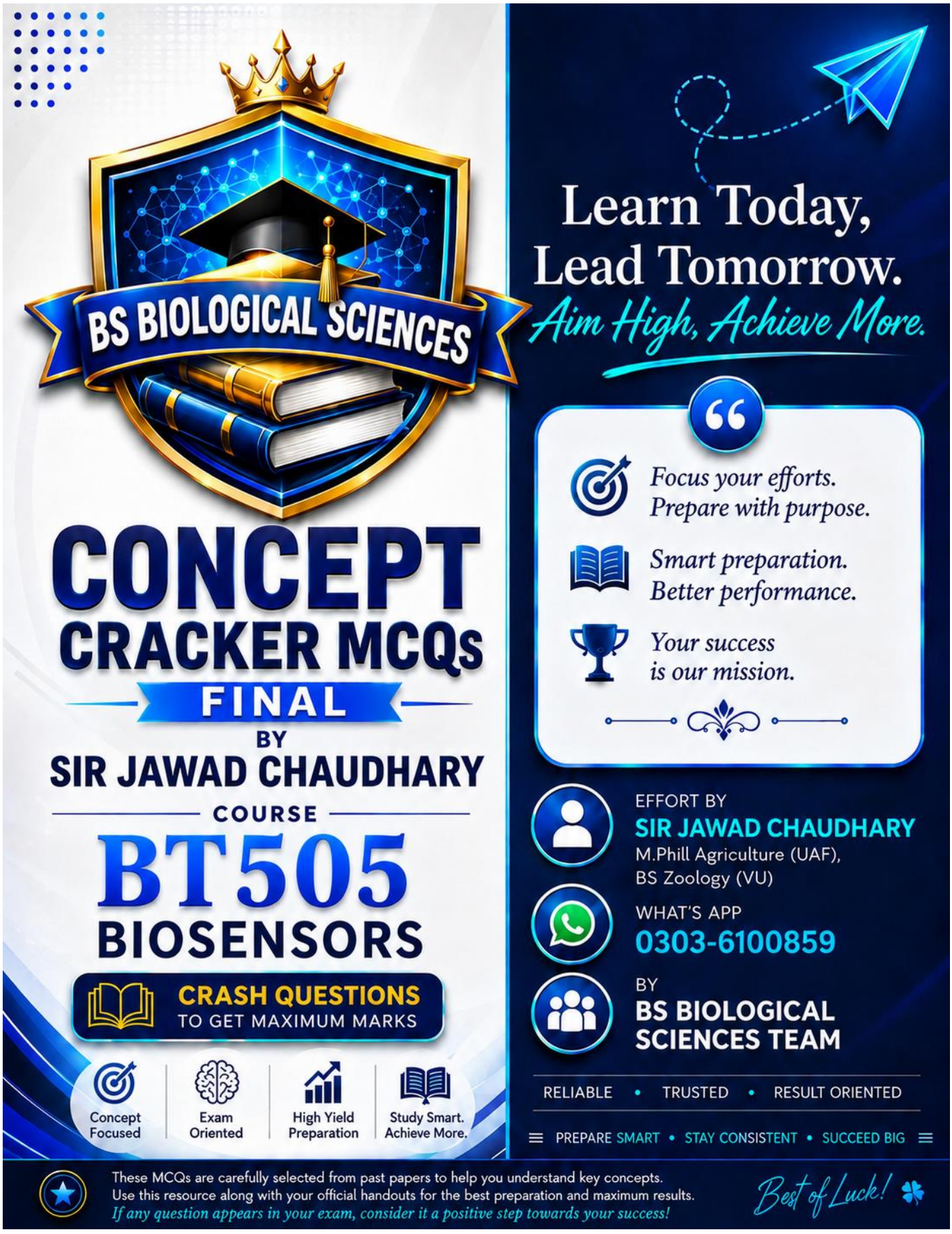


JAWAD CHAUDHARY

MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN BT505 BIOSENSORS CONCEPT CRACKER MCQS FINAL EXAMS

Main Group (BS Biological Sciences) *https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN*

BT505 Biosensors *https://chat.whatsapp.com/HPifmMOikXPC6tu2PWn7Yu*

Q1: A biosensor measures biological or chemical reactions by generating signals proportional to the concentration of an _____.

Answer: Analyte

Q2: Glucose in a glucose biosensor is an example of an _____.

Answer: Analyte

Q3: The ability of a biosensor to detect a specific analyte in the presence of contaminants is called _____.

Answer: Selectivity

Q4: The interaction between an antigen and an antibody best demonstrates biosensor _____.

Answer: Selectivity

Q5: The ability of a biosensor to produce identical responses under duplicated experimental conditions is called _____.

Answer: Reproducibility

Q6: Reproducibility depends mainly on the precision and _____ of the biosensor.

Answer: Accuracy

Q7: The ability of a biosensor to provide similar results every time a sample is measured is called _____.

Answer: Precision

Q8: The ability of a biosensor to produce results close to the true value is known as _____.

Answer: Accuracy

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Q9: The degree of susceptibility of a biosensor to environmental disturbances is called _____.

Answer: Stability

Q10: High affinity between analyte and bioreceptor improves biosensor _____.

Answer: Stability

Q11: The minimum amount of analyte that can be detected by a biosensor is called the _____.

Answer: Limit of Detection (LOD)

Q12: Another term commonly used for the limit of detection (LOD) is _____.

Answer: Sensitivity

Q13: The relationship between biosensor output and analyte concentration is represented by the equation **y = mc**, where **m** represents _____.

Answer: Sensitivity

Q14: The smallest change in analyte concentration that produces a detectable signal change is called _____.

Answer: Resolution

Q15: The range of analyte concentrations over which biosensor response remains linear is called the _____.

Answer: Linear range

Q16: A molecule that specifically recognizes an analyte is called a _____.

Answer: Bioreceptor

Q17: Enzymes, antibodies, DNA and aptamers are examples of _____.

Answer: Bioreceptors

Q18: Signal generation after interaction between analyte and bioreceptor is known as _____.

Answer: Biorecognition

Q19: The component that converts a biorecognition event into a measurable signal is the _____.

Answer: Transducer

Q20: The process of converting one form of energy into another is known as _____.

Answer: Signalisation

Q21: Most biosensor transducers generate either optical or _____ signals.

Answer: Electrical

Q22: The electronics section of a biosensor performs signal _____ before display.

Answer: Conditioning

Q23: The conversion of analogue signals into digital form is performed by the biosensor _____.

Answer: Electronics

Q24: The user-friendly presentation of biosensor results is the function of the _____.

Answer: Display

Q25: MEMS stands for Micro-Electro-Mechanical _____.

Answer: Systems

Q26: MEMS devices are manufactured using _____ techniques.

Answer: Microfabrication

Q27: The main criterion for a MEMS device is the presence of some form of _____ functionality.

Answer: Mechanical

Q28: Microsensors and micro actuators are classified as _____.

Answer: Transducers

Q29: A microsensor usually converts a mechanical signal into an _____ signal.

Answer: Electrical

Q30: Biosensors that measure current generated during enzyme-catalyzed redox reactions are called _____ biosensors.

Answer: Amperometric

Q31: In an amperometric biosensor, substrate concentration is directly proportional to the magnitude of the _____.

Answer: Current

Q32: The most common example of an amperometric biosensor is the _____ biosensor.

Answer: Blood glucose

Q33: Second-generation amperometric biosensors use a _____ to transfer electrons to the electrode.

Answer: Mediator

Q34: Blood glucose biosensors are widely used by _____ patients.

Answer: Diabetic

Q35: Potentiometric biosensors determine changes in _____ concentration.

Answer: Ionic

Q36: The most commonly used electrode in potentiometric biosensors is the _____ electrode.

Answer: pH

Q37: The potential difference in a potentiometric biosensor is directly proportional to the _____ concentration.

Answer: Substrate

Q38: ISFET stands for Ion-Selective _____ Effect Transistor.

Answer: Field

Q39: Conductometric biosensors measure changes in electrical _____ caused by ionic species.

Answer: Conductivity

Q40: A conductometric biosensor commonly used during dialysis measures _____ concentration.

Answer: Urea

Q41: Thermometric biosensors operate on the principle of biological _____ production.

Answer: Heat

Q42: Optical biosensors commonly utilize fluorescence, absorbance and other _____ measurements.

Answer: Optical

Q43: Optical biosensors are suitable for non-electrical _____ sensing.

Answer: Remote

Q44: The fiber optic lactate biosensor measures changes in _____ concentration.

Answer: Oxygen

Q45: Increased lactate concentration results in increased _____ output in the fiber optic lactate biosensor.

Answer: Fluorescence

Q46: Optical glucose biosensors commonly use paper strips containing glucose oxidase, horseradish peroxidase and a _____.

Answer: Chromogen

Q47: Piezoelectric biosensors are also known as _____ biosensors.

Answer: Acoustic

Q48: Piezoelectric biosensors are based on the vibration frequency of _____ crystals.

Answer: Piezoelectric

Q49: Immunobiosensors operate on the principle of immunological _____.

Answer: Specificity

Q50: In immunobiosensors, an immobilized antibody directly binds with the _____.

Answer: Antigen

Q51: MEMS components are commonly manufactured on _____ wafers.

Answer: Silicon

Q52: Improved methods for _____ and texturing of silicon wafers enabled MEMS fabrication.

Answer: Etching

Q53: Dry etching is also known as _____ etching.

Answer: Plasma

Q54: During plasma etching, highly reactive radicals react with _____ to remove material.

Answer: Silicon

Q55: Deep trenches in MEMS structures are generally produced using _____ etching.

Answer: Plasma

Q56: PECVD stands for Plasma Enhanced Chemical Vapor _____.

Answer: Deposition

Q57: LPCVD stands for Low Pressure Chemical Vapor _____.

Answer: Deposition

Q58: Compared to PECVD, LPCVD operates at relatively _____ temperatures.

Answer: High

Q59: Common dielectric layers deposited during MEMS fabrication include silicon oxide and silicon _____.

Answer: Nitride

Q60: Consumer electronics, biomedical devices, and aerospace systems are important _____ of MEMS.

Answer: Applications

Q61: MEMS pressure sensors were first widely used for measuring _____ pressure.

Answer: Blood

Q62: Normal intraocular pressure (IOP) ranges from _____ to _____ mmHg.

Answer: 10 to 22

Q63: An intraocular pressure above _____ mmHg is considered a major risk factor for glaucoma.

Answer: 22

Q64: Continuous monitoring of IOP in glaucoma patients is performed using implantable _____ pressure sensors.

Answer: MEMS

Q65: Modern pacemakers adjust heart rate using MEMS _____.

Answer: Accelerometers

Q66: Implantable defibrillators restore a normal heart rhythm by delivering electrical _____.

Answer: Shocks

Q67: MEMS inertial sensors include accelerometers and _____.

Answer: Gyroscopes

Q68: The iBOT mobility system balances on _____ wheels.

Answer: Two

Q69: Miniaturization of hearing aids using MEMS reduces size, cost, and _____ consumption.

Answer: Power

Q70: Microfluidics deals with handling very small volumes of _____.

Answer: Fluids

Q71: Point-of-care diagnosis is commonly abbreviated as _____.

Answer: POC

Q72: The first commercial microfluidic pregnancy test device was _____.

Answer: ClearBlue™

Q73: The Alere Pima™ system measures the _____ count in HIV patients.

Answer: CD4

Q74: The Alere Pima™ CD4 test requires approximately _____ µL of whole blood.

Answer: 25

Q75: JewelPUMP™ is a MEMS-based system designed for continuous _____ delivery.

Answer: Insulin

Q76: MEMS-based microneedles are generally shorter than _____ mm.

Answer: 1

Q77: Microneedles are commonly used for drug delivery and _____ sampling.

Answer: Fluid

Q78: Minimally Invasive Surgery is commonly abbreviated as _____.

Answer: MIS

Q79: One major advantage of MIS is reduced _____ time.

Answer: Recovery

Q80: Chemical sensors and inertial sensors are major categories of _____ sensors.

Answer: MEMS

Q81: An Ion Selective Electrode measures the _____ of a specific ion in solution.

Answer: Potential

Q82: The pH electrode is an example of an _____.

Answer: Ion Selective Electrode (ISE)

Q83: The electric potential generated by an ISE is measured against a _____ electrode.

Answer: Reference

Q84: The potential difference in an ISE depends on the _____ of the selected ion.

Answer: Activity

Q85: The basic galvanic cell equation is Ecell = EISE − _____.

Answer: Eref

Q86: The simplest method of determining ion concentration with an ISE is called direct _____.

Answer: Calibration

Q87: The addition of known amounts of a standard solution is called standard _____.

Answer: Additions

Q88: During titration using an ISE, the ion concentration is calculated from the volume of the _____ used.

Answer: Titrant

Q89: The selectivity of an ion-selective electrode is mainly determined by the _____ material.

Answer: Membrane

Q90: Polymer membrane electrodes commonly used _____ as an inert matrix.

Answer: PVC

Q91: Solid-state electrodes utilize relatively insoluble inorganic _____.

Answer: Salts

Q92: Gas sensing electrodes contain a gas-permeable membrane and an internal _____ solution.

Answer: Buffer

Q93: Glass membrane electrodes are most commonly used for measuring _____.

Answer: pH

Q94: Accurate ISE measurements require proper _____ of the solution.

Answer: Agitation

Q95: Before storing, an ion-selective electrode should be cleaned and properly _____.

Answer: Conditioned

Q96: At least _____ minutes should be allowed for ISE readings to stabilize.

Answer: 15

Q97: A calibration curve is generally established using _____ standard solutions.

Answer: Three

Q98: Between measurements, the ISE should be rinsed with _____ or distilled water.

Answer: Deionized

Q99: Polymer membrane electrodes should never come into contact with organic _____.

Answer: Solvents

Q100: Gas biosensors reviewed in the handouts are capable of detecting volatile compounds present in the _____ phase.

Answer: Gas

Q101: Gas biosensors are designed to detect vapors of volatile _____.

Answer: Compounds

Q102: Sulphite oxidase is an enzyme used in the detection of _____ vapors.

Answer: Sulfur dioxide (SO_2)

Q103: Alcohol oxidase is employed in gas biosensors for detecting _____ vapors.

Answer: Alcohol

Q104: Formaldehyde dehydrogenase is used in gas biosensors for detecting _____.

Answer: Formaldehyde

Q105: Polyphenol oxidase is used for detecting _____ compounds.

Answer: Phenolic

Q106: The enzyme used in a gas biosensor for phenol detection is _____.

Answer: Tyrosinase

Q107: Gas biosensors discussed in the handouts can detect phenol vapors in the _____ range.

Answer: ppb

Q108: The key component responsible for selective recognition in a biosensor is the biological _____.

Answer: Molecule

Q109: Biomolecules interact _____ with their target compounds in a biosensor.

Answer: Selectively

Q110: The transduction stage is mainly responsible for signal _____.

Answer: Amplification

Q111: One type of biosensor transduction is _____ transduction, which measures electrical current.

Answer: Amperometric

Q112: Piezoelectric biosensors detect analytes through changes in vibration _____.

Answer: Frequency

Q113: ISFET stands for Ion-Sensitive _____ Effect Transistor.

Answer: Field

Q114: In an ISFET, the solution acts as the _____ electrode.

Answer: Gate

Q115: The current through an ISFET changes according to the concentration of _____.

Answer: Ions

Q116: Common gate material used in ISFETs is silicon _____.

Answer: Dioxide (SiO_2)

Q117: The threshold voltage of an ISFET depends on the _____ of the solution.

Answer: pH

Q118: The oxide surface charge in an ISFET is explained by the _____ binding model.

Answer: Site

Q119: The hydroxyl groups on the oxide surface behave in an _____ manner.

Answer: Amphoteric

Q120: X-rays belong to the _____ spectrum.

Answer: Electromagnetic

Q121: Visible light is only one portion of the _____ spectrum.

Answer: Electromagnetic

Q122: Night vision goggles primarily detect _____ radiation.

Answer: Infrared

Q123: Skin tanning is mainly caused by _____ radiation.

Answer: Ultraviolet

Q124: Dentists commonly use _____ rays to image teeth.

Answer: X

Q125: Gamma rays are widely used for _____ imaging.

Answer: Medical

Q126: X-rays were discovered by Wilhelm Conrad _____.

Answer: Röntgen

Q127: X-rays are produced when high-speed electrons strike a _____ target.

Answer: Metal

Q128: Electrons used to produce X-rays originate from a heated _____.

Answer: Filament

Q129: Bones appear _____ on X-ray images because they absorb more radiation.

Answer: White

Q130: Soft tissues appear _____ on X-ray images because more X-rays pass through them.

Answer: Dark

Q131: X-rays can pass through most _____ tissues.

Answer: Soft

Q132: A Field Effect Transistor is commonly abbreviated as _____.

Answer: FET

Q133: Current in a FET flow through a semiconductor _____.

Answer: Channel

Q134: The three main terminals of a FET are source, drain, and _____.

Answer: Gate

Q135: A small change in gate voltage produces a large change in _____ current.

Answer: Drain

Q136: The two major classifications of FET are JFET and _____.

Answer: MOSFET

Q137: MOSFET stands for Metal Oxide Semiconductor _____ Effect Transistor.

Answer: Field

Q138: In a MOSFET, the gate is electrically insulated by a thin _____ layer.

Answer: Oxide

Q139: MOSFETs possess extremely high input _____.

Answer: Impedance

Q140: MOSFETs are highly susceptible to _____ discharge damage.

Answer: Electrostatic

Q141: FETs are generally preferred for _____ signal amplification.

Answer: Weak

Q142: Thermal biosensors are also known as enzyme _____.

Answer: Thermistors

Q143: Thermal biosensors detect heat produced during _____ reactions.

Answer: Enzymatic

Q144: Split-flow thermal biosensors use a reference column without the active _____.

Answer: Enzyme

Q145: Clinical chemistry is one of the major application areas of _____ biosensors.

Answer: Thermal

Q146: Environmental monitoring using thermal biosensors is referred to as _____-guard.

Answer: Toxi

Q147: A MEMS differential thermal biosensor detects heat using a _____.

Answer: Thermopile

Q148: Enzymes in MEMS thermal biosensors are immobilized on _____.

Answer: Microbeads

Q149: The MEMS thermal biosensor described in the handout was experimentally tested using _____ solutions.

Answer: Glucose

Q150: Optical biosensors convert biochemical interactions into _____ output signals.

Answer: Optical

Q151: Optical biosensors utilize _____ principles to convert biochemical interactions into measurable signals.

Answer: Optical

Q152: Optical biosensors are considered an alternative to conventional _____ techniques.

Answer: Analytical

Q153: One major advantage of optical biosensors is their high _____.

Answer: Sensitivity

Q154: Optical biosensors are capable of _____-free detection of analytes.

Answer: Label

Q155: The component responsible for recognizing a specific analyte is called the _____ element.

Answer: Recognition

Q156: Enzymes, antibodies, receptors, DNA, and cells are examples of _____ elements.

Answer: Biorecognition

Q157: The sampling unit introduces the _____ into the detector.

Answer: Analyte

Q158: The recognition element provides biodetection _____.

Answer: Specificity

Q159: The glucose biosensor described in the handout provides real-time monitoring of glucose concentration in the _____ of rodents.

Answer: Brain

Q160: The glucose biosensor rejects common electroactive _____.

Answer: Interferents

Q161: The glucose biosensor can continuously monitor glucose concentration for more than _____ hours.

Answer: 96

Q162: Optical biosensors mainly monitor changes in light absorption between reactants and _____.

Answer: Products

Q163: Colorimetric test strips are generally _____-use devices.

Answer: Single

Q164: Glucose oxidase is commonly used in optical biosensors for _____ monitoring.

Answer: Blood glucose

Q165: Horseradish peroxidase catalyzes the oxidation of the _____.

Answer: Chromogen

Q166: Hydrogen peroxide oxidizes the weakly colored chromogen into a highly colored _____.

Answer: Dye

Q167: Colored glucose strips are commonly analyzed using portable _____ meters.

Answer: Reflectance

Q168: Firefly luciferase-based biosensors detect the presence of _____.

Answer: Bacteria

Q169: The amount of ATP released is approximately proportional to the number of _____ present.

Answer: Bacteria

Q170: Firefly luciferase catalyzes a reaction that produces _____ light.

Answer: Yellow

Q171: The wavelength of light produced by the luciferase reaction is _____ nm.

Answer: 562

Q172: Light generated in luciferase biosensors is detected using a _____ tube or photodiode.

Answer: Photomultiplier

Q173: Immobilization of luciferase mainly reduces the _____ of analysis.

Answer: Cost

Q174: A biosignal is a description of a _____ phenomenon.

Answer: Physiological

Q175: Biosignals provide information about the _____ or behavior of a living object.

Answer: State

Q176: ECG and EEG are examples of _____ signals.

Answer: Biosignals

Q177: Biosignals help understand the underlying _____ mechanisms of biological events.

Answer: Physiological

Q178: According to physical nature, biosignals may be electrical, magnetic, chemical, mechanical, optical, or _____.

Answer: Thermal

Q179: Electrical biosignals are generated due to intra- and extracellular _____ currents.

Answer: Ionic

Q180: EEG stands for Electroencephalo_____.

Answer: Gram

Q181: ECG stands for Electrocardio_____.

Answer: Gram

Q182: EMG stands for Electromyo_____.

Answer: Gram

Q183: EOG stands for Electrooculo_____.

Answer: Gram

Q184: GSR stands for Galvanic Skin _____.

Answer: Response

Q185: MEG stands for Magnetoencephalo_____.

Answer: Graphy

Q186: MEG measures the magnetic fields produced by _____ activity.

Answer: Neuronal

Q187: MEG provides much higher _____ resolution than fMRI.

Answer: Temporal

Q188: MCG is used to measure magnetic fields generated by the _____.

Answer: Heart

Q189: Blood glucose level is an example of a _____ biosignal.

Answer: Chemical

Q190: Blood oxygen level is another example of a _____ biosignal.

Answer: Chemical

Q191: Blood pressure is classified as a _____ biosignal.

Answer: Mechanical

Q192: Accelerometer signals describing body movement are _____ biosignals.

Answer: Mechanical

Q193: Acoustic biosignals are a subset of _____ biosignals.

Answer: Mechanical

Q194: Heart sounds are recorded using _____.

Answer: Phonocardiography

Q195: A phonocardiogram mainly reflects sounds produced by _____ valve closures.

Answer: Heart

Q196: Respiratory sounds include normal breathing sounds and _____.

Answer: Crackles

Q197: Body temperature maps are examples of _____ biosignals.

Answer: Thermal

Q198: Hormone-related signals originate from the _____ system.

Answer: Endocrine

Q199: EEG and MEG are signals associated with the _____ system.

Answer: Nervous

Q200: ECG, MCG, and blood pressure measurements are associated with the _____ system.

Answer: Cardiovascular

Q201: Electroretinography measures the electrical responses of the _____.

Answer: Retina

Q202: ERG primarily records the activity of rods, cones, bipolar cells, amacrine cells, and _____ cells.

Answer: Ganglion

Q203: Visual Evoked Potentials are commonly abbreviated as _____.

Answer: VEP

Q204: Audiometry is associated with the _____ system.

Answer: Auditory

Q205: Spirometry is used to evaluate the _____ system.

Answer: Respiratory

Q206: Plethysmography is mainly used to measure lung _____.

Answer: Volume

Q207: Electrogastrogram is related to the _____ system.

Answer: Gastrointestinal

Q208: The most important criterion when selecting a redox enzyme is its _____ toward the analyte.

Answer: Selectivity

Q209: The primary redox center should lie within a suitable _____ window.

Answer: Potential

Q210: A biosensor enzyme should remain stable during operation and _____ conditions.

Answer: Storage

Q211: Enzymes exhibit very high catalytic _____.

Answer: Activity

Q212: The catalytic turnover number of an enzyme is represented by _____.

Answer: kcat

Q213: High substrate specificity is a major advantage of using _____ in biosensors.

Answer: Enzymes

Q214: Enzyme-based biosensors usually operate at relatively _____ potentials.

Answer: Moderate

Q215: Immobilization often improves enzyme _____.

Answer: Stability

Q216: One disadvantage of enzymes is their relatively large molecular _____.

Answer: Size

Q217: The active site of many enzymes is deeply buried within the protein _____.

Answer: Shell

Q218: Artificial redox mediators are often required because direct _____ transfer is difficult.

Answer: Electron

Q219: The long-term stability of enzyme biosensors is limited by enzyme _____.

Answer: Lifetime

Q220: Operational lifetimes of many enzyme biosensors range from _____ to _____ days.

Answer: 30 to 60

Q221: First-generation biosensors directly detect enzymatically produced _____ or consumed substrates.

Answer: Products

Q222: In first-generation glucose biosensors, glucose oxidase is commonly abbreviated as _____.

Answer: GOx

Q223: Glucose concentration can be monitored by measuring the increase in _____ concentration.

Answer: Hydrogen peroxide (H_2O_2)

Q224: Another first-generation approach monitors the decrease in _____ concentration.

Answer: Oxygen (O_2)

Q225: First-generation biosensors require the monitored compound to be _____ active.

Answer: Electrochemically

Q226: Glucose oxidase acts as the biological _____ element in first-generation glucose biosensors.

Answer: Recognition

Q227: Oxygen serves as a natural redox _____ in glucose oxidase reactions.

Answer: Mediator

Q228: Amperometric biosensors convert chemical reactions into measurable _____.

Answer: Current

Q229: The electrical signal generated in an amperometric biosensor is proportional to analyte _____.

Answer: Concentration

Q230: A major commercial application of amperometric biosensors is _____ monitoring.

Answer: Blood glucose

Q231: In biosensor terminology, the analyte is the substance to be _____.

Answer: Detected

Q232: The interaction between analyte and bioreceptor is known as bio-_____.

Answer: Recognition

Q233: A transducer converts one form of _____ into another.

Answer: Energy

Q234: The display unit presents biosensor results in a _____-friendly manner.

Answer: User

Q235: High selectivity reduces interference from unwanted _____.

Answer: Contaminants

Q236: Sensitivity is commonly expressed by the biosensor's limit of _____.

Answer: Detection

Q237: A biosensor with good reproducibility provides highly _____ responses.

Answer: Consistent

Q238: Stability is particularly important during long _____ periods.

Answer: Incubation

Q239: The response of transducers may be affected by _____ changes.

Answer: Temperature

Q240: The affinity of a bioreceptor determines the strength of analyte _____.

Answer: Binding

Q241: Resolution refers to the smallest detectable change in analyte _____.

Answer: Concentration

Q242: Linearity describes the relationship between analyte concentration and biosensor _____.

Answer: Response

Q243: The linear range represents concentrations over which the response remains _____.

Answer: Linear

Q244: Electronics convert analog signals into _____ signals.

Answer: Digital

Q245: Signal amplification is performed before information reaches the _____ unit.

Answer: Display

Q246: Optical and electrical signals are the most common outputs generated by _____.

Answer: Transducers

Q247: MEMS technology combines mechanical components with _____ circuits.

Answer: Electronic

Q248: Microsensors are classified as energy-converting _____.

Answer: Transducers

Q249: Biosensors are widely applied in disease monitoring and _____ discovery.

Answer: Drug

Q250: The overall performance of a biosensor depends on proper optimization of its static and _____ attributes.

Answer: Dynamic

Q251: The primary purpose of a biosensor is to detect the concentration of an _____.

Answer: Analyte

Q252: Disease-causing microorganisms can be detected using _____.

Answer: Biosensors

Q253: Drug discovery is one of the important applications of _____.

Answer: Biosensors

Q254: The first consideration when selecting a bioreceptor is its _____.

Answer: Selectivity

Q255: Antibodies are classically immobilized on the surface of the _____.

Answer: Transducer

Q256: Ambient disturbances mainly affect the _____ of a biosensor.

Answer: Stability

Q257: High-affinity bioreceptors encourage strong electrostatic or _____ bonding.

Answer: Covalent

Q258: A prostate-specific antigen (PSA) concentration of _____ ng/mL in blood is associated with prostate cancer.

Answer: 4

Q259: The slope (m) in the linearity equation **y = mc** represents the biosensor _____.

Answer: Sensitivity

Q260: The process of converting analog signals into digital form is performed by the biosensor _____.

Answer: Electronics

Q261: The output of a biosensor may be displayed in numeric, graphic, tabular, or _____ form.

Answer: Image

Q262: MEMS devices may range in size from below one micron to several _____.

Answer: Millimeters

Q263: Microactuators are classified as energy-converting _____.

Answer: Transducers

Q264: In first-generation amperometric biosensors, electron transfer occurs _____ from the enzyme to the electrode.

Answer: Directly

Q265: Second-generation amperometric biosensors improve electron transfer by using a _____.

Answer: Mediator

Q266: Potentiometric biosensors are particularly sensitive to changes in _____ concentration.

Answer: Ionic

Q267: Conductometric biosensors detect changes in electrical _____.

Answer: Conductivity

Q268: Biological reactions producing heat form the basis of _____ biosensors.

Answer: Thermometric

Q269: Fiber optic biosensors commonly utilize changes in _____ intensity.

Answer: Fluorescence

Q270: Piezoelectric biosensors detect analytes through changes in crystal vibration _____.

Answer: Frequency

Q271: Immunobiosensors rely on antigen–antibody _____.

Answer: Interaction

Q272: Plasma etching removes silicon through reactions with highly reactive _____.

Answer: Radicals

Q273: LPCVD deposits layers mainly through _____ reactions.

Answer: Thermal

Q274: Microfluidic systems manipulate very small volumes of _____.

Answer: Fluids

Q275: Continuous glucose delivery systems such as JewelPUMP™ employ a MEMS _____.

Answer: Nanopump

Q276: Ion-selective electrodes determine ion concentration by measuring electrical _____.

Answer: Potential

Q277: Gas sensing electrodes contain a gas-permeable membrane and an internal buffer _____.

Answer: Solution

Q278: Polymer membrane electrodes are commonly sealed to the end of a _____ tube.

Answer: PVC

Q279: The most common glass membrane electrode is the _____ electrode.

Answer: pH

Q280: Ion-selective electrodes should remain _____ when not in use.

Answer: Moist

Q281: Phenol gas biosensors use the enzyme _____.

Answer: Tyrosinase

Q282: ISFET threshold voltage changes according to the _____ of the test solution.

Answer: pH

Q283: X-rays are produced when accelerated electrons strike a _____ target.

Answer: Metal

Q284: Bones absorb more X-rays than soft tissues because they are more _____.

Answer: Dense

Q285: The control electrode of a FET is called the _____.

Answer: Gate

Q286: MOSFETs have extremely high input _____ because of the insulating oxide layer.

Answer: Impedance

Q287: Thermal biosensors detect heat released during _____ reactions.

Answer: Enzymatic

Q288: Optical biosensors provide direct, real-time, and _____-free detection.

Answer: Label

Q289: Firefly luciferase biosensors detect bacteria by measuring released _____.

Answer: ATP

Q290: EEG is classified as an _____ bio signal.

Answer: Electrical

Q291: MEG measures magnetic fields generated by _____ activity.

Answer: Neuronal

Q292: Blood pressure is classified as a _____ bio signal.

Answer: Mechanical

Q293: Heart sounds are recorded using _____.

Answer: Phonocardiography

Q294: Body temperature distribution is an example of a _____ bio signal.

Answer: Thermal

Q295: A major advantage of enzyme-based biosensors is their high catalytic _____.

Answer: Activity

Q296: Artificial redox mediators are used when direct _____ transfer is difficult.

Answer: Electron

Q297: First-generation glucose biosensors commonly use _____ oxidase as the recognition enzyme.

Answer: Glucose

Q298: The electrochemical detection of hydrogen peroxide is a characteristic of _____-generation biosensors.

Answer: First

Q299: The overall response of a biosensor depends on the interaction between the analyte and the _____.

Answer: Bioreceptor

Q300: The four basic components of a biosensor are bioreceptor, transducer, electronics, and _____.

Answer: Display

🚨 Concept Cracker MCQS Most Important Repeated Conceptual 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

BS Biological Sciences Team 🧬🎓

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